

Paper 2

1. Evaluate $\int_0^2 \frac{3}{7x+2} dx$.

$$\begin{aligned}
 \int_0^2 \frac{3}{7x+2} dx &= \frac{3}{7} \int_0^2 \frac{7}{7x+2} dx \\
 &= \frac{3}{7} [\ln|7x+2|]_0^2 \quad \checkmark \text{ 1 mark} \\
 &= \frac{3}{7} [\ln|7 \times 2 + 2| - \ln|7 \times 0 + 2|] \\
 &= \frac{3}{7} [\ln 16 - \ln 2] \\
 &= \frac{3}{7} [4 \ln 2 - \ln 2] \\
 &= \frac{3}{7} \times 3 \ln 2 \\
 &= \frac{9}{7} \ln 2 \quad \checkmark \text{ 1 mark} \\
 &\quad \text{OR } \frac{3}{7} \ln 8
 \end{aligned}$$

ks)

2. Find $\int (\sin 5x - \frac{3}{e^{2x}}) dx$.

$$\begin{aligned}
 &\int (\sin 5x - \frac{3}{e^{2x}}) dx \\
 &= \int [-\frac{1}{5}(-5 \sin 5x) - 3e^{-2x}] dx \\
 &= \int [-\frac{1}{5}(-5 \sin 5x) - \frac{3 \times (-2)}{(-2)} e^{-2x}] dx \\
 &= -\frac{1}{5} \cos 5x + \frac{3}{2} e^{-2x} + C \quad \checkmark \text{ 1 mark for correct integration}
 \end{aligned}$$

3. Use three applications of the trapezoidal rule to estimate the value of $\int_1^4 \sqrt{x-1} dx$. Give your answer correct to 4 significant figures.

x	1	2	3	4
$f(x)$	0	1	$\sqrt{2}$	$\sqrt{3}$

$$\begin{aligned}
 \int_a^b f(x) dx &\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + f(x_2)]\} \\
 \int_1^4 \sqrt{x-1} dx &\approx \frac{4-1}{2 \times 3} \{0 + \sqrt{3} + 2(1 + \sqrt{2})\} \\
 &\approx 3.280238966.. \\
 &\approx 3.280 \text{ (4 significant figures)}
 \end{aligned}$$

4. Solve $\sin x = -\frac{1}{2}$, where $0 \leq x \leq 2\pi$. Giving the answer in exact form in radians

$$x = \frac{-\pi}{6}$$

$$= \frac{11\pi}{6} \text{ and } \frac{7\pi}{6}$$

5. Find $\int 10^{5x-1} dx$.

$$= \frac{10^{5x-1}}{5 \ln 10} + C$$

6. A group of medical researchers conducted an investigation to study the impact of different dosages of a new drug on the duration of relief from an allergic reaction.

An experiment was carried out using a random sample of seven patients. The data collected is recorded in the table.

<i>Dosage (mL)</i>	3	4	4	5	7	7	9
<i>Duration of relief (hours)</i>	5	9	12	9	18	22	22

- a) Find the equation of the least-squares regression line. Give your answer in the form $y = Ax + B$, where A & B are correct to 2 decimal places.

$$A = -2.54 \quad B = 2.94$$

$$y = 2.94x - 2.54$$

- b) By finding Pearson's correlation coefficient, describe the strength and direction of the relationship between the dosage and the duration of relief.

$$r = 0.93 \quad \textcircled{1}$$

As r is positive and close to 1 the relationship is strong positive $\textcircled{1}$

- c) Using the equation found in part (a), estimate the hours of relief with a dosage of 4.3 mL. Give your answer correct to the nearest hour.

$$y = 2.94 \times 4.3 - 2.54$$

$$= 10 \text{ hours} \quad \textcircled{1}$$

7. Evaluate in simplest form $\int_0^{\frac{\pi}{2}} \sin 2x \, dx$.

$$\left[-\frac{1}{2} \cos 2x \right]_0^{\pi/2}$$

$$= -\frac{1}{2} \cos \pi - -\frac{1}{2} \cos 0$$

$$= 1$$

8. Differentiate the following with respects to x .

a. $\frac{3x+2}{x+4}$

$$u = 3x+2 \quad v = x+4$$

$$u' = 3 \quad v' = 1$$

$$\frac{d}{dx} = \frac{3(x+4) - (3x+2)}{(x+4)^2}$$

$$= \frac{3x+12-3x-2}{(x+4)^2}$$

$$= \frac{10}{(x+4)^2}$$

b. $x(1+2x)^4$

$$u = x \quad v = (1+2x)^4$$

$$u' = 1 \quad v' = 8(1+2x)^3$$

$$\frac{d}{dx} = 8x(1+2x)^3 + (1+2x)^4$$

$$= (1+2x)^3 (8x+1+2x)$$

$$= (10x+1)(1+2x)^3$$

9. A discrete random variable X has the probability distribution table shown below.

$X = x$	5	10	15	20
$P(x)$	0.1	0.25	n	0.35

- a) Find the value of n .

$$0.1 + 0.25 + n + 0.35 = 1$$

$$n = 0.3 \quad \checkmark$$

- b) Find the expected value $E(X)$.

$$E(X) = \sum x p$$

$$= 5(0.1) + 10(0.25) + 15(0.3) + 20(0.35)$$

$$= 14.5 \quad \checkmark$$

10. Differentiate with respect to x .

- a) $y = (3x^2 - 7)^6$.

$$y' = 6(3x^2 - 7)^5 \cdot (6x)$$

$$= 36x(3x^2 - 7)^5 \quad \checkmark$$

- b) $y = \frac{2x-1}{x+8}$.

$$u = 2x - 1, \quad v = x + 8$$

$$u' = 2, \quad v' = 1$$

$$y' = \frac{2(x+8) - (2x-1)}{(x+8)^2}$$

$$= \frac{2x + 16 - 2x + 1}{(x+8)^2}$$

$$= \frac{17}{(x+8)^2}$$

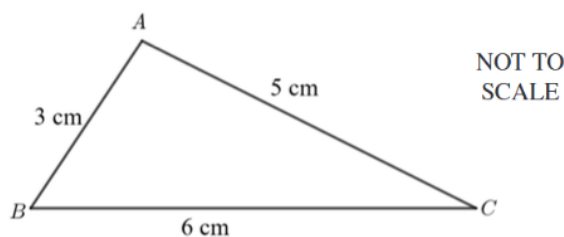
11. Find the equation of the tangent to the curve $y = x^3 + \ln x + 5$ at the point (1, 6).

$$\begin{aligned}
 y &= x^3 + \ln x + 5 \\
 y' &= 3x^2 + \frac{1}{x} \quad \checkmark \\
 \text{at } x=1, \quad m &= 3(1)^2 + \frac{1}{1} \\
 m &= 4 \quad \checkmark \\
 y - y_1 &= m(x - x_1) \\
 y - 6 &= 4(x - 1) \\
 y &= 4x + 2 \quad \checkmark
 \end{aligned}$$

12. Solve the equation $\sqrt{3} + 2\cos\theta = 0$ in the domain $0 \leq \theta \leq 2\pi$

$$\begin{aligned}
 2\cos\theta &= -\sqrt{3} \\
 \cos\theta &= -\frac{\sqrt{3}}{2} \\
 \text{related angle: } \theta &= \frac{\pi}{6} \text{ rad} \\
 \therefore \theta &= \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6} \\
 &= \frac{5\pi}{6}, \frac{7\pi}{6}
 \end{aligned}$$

13. ABC is a triangle in which $AB = 3$ cm, $BC = 6$ cm and $CA = 5$ cm as shown in the diagram below.



Use the Cosine rule to calculate $\angle BAC$ to the nearest degree

$$\begin{aligned}
 \cos A &= \frac{3^2 + 5^2 - 6^2}{2(3)(5)} \\
 &= \frac{-1}{15} \\
 \angle A &= \cos^{-1}\left(\frac{-1}{15}\right) \\
 &= 93.8^\circ \\
 &= 94^\circ \text{ (nearest degree)}
 \end{aligned}$$

14. Consider the curve $y = x^3 - 12x^2 + 36x$.

a) Find the x and y intercepts.

y intercept ($x = 0$) is 0

x intercept ($y = 0$)

$$\begin{aligned}y &= x^3 - 12x^2 + 36x \\0 &= x(x^2 - 12x + 36) \\x(x - 6)^2 &= 0\end{aligned}$$

$\therefore x = 0$ or $x = 6$ (x intercepts are 0 and 6)

b) Find the location and nature of any stationary points.

$$\begin{aligned}y &= x^3 - 12x^2 + 36x \\y' &= 3x^2 - 24x + 36 \\y'' &= 6x - 24\end{aligned}$$

Stationary points occur when $y' = 0$

$$\begin{aligned}3x^2 - 24x + 36 &= 0 \\x^2 - 8x + 12 &= 0 \\(x - 2)(x - 6) &= 0 \\\therefore x &= 2 \text{ or } x = 6\end{aligned}$$

For $x = 2$

$$\begin{aligned}y &= 2^3 - 12(2)^2 + 36(2) \\&= 32 \\y'' &= 6(2) - 24 \\&= -12 < 0 \rightarrow \text{conc down}\end{aligned}$$

For $x = 6$

$$\begin{aligned}y &= 6^3 - 12(6)^2 + 36(6) \\&= 0 \\y'' &= 6(6) - 24 \\&= 12 > 0 \rightarrow \text{conc up}\end{aligned}$$

$\therefore$ Stationary points at $(2, 32)$ and $(6, 0)$ are a local maximum and a local minimum, respectively.

c) Find the point of inflection.

Possible inflection when $y'' = 0$

$$\begin{aligned}6x - 24 &= 0 \\x &= 4\end{aligned}$$

x	3	4	5
y''	-4 < 0	0	6 > 0

When $x = 4$, $y = 16$

As concavity changes across the point, $(4, 16)$ is a point of inflection